

HEATING, VENTILATION, AND AIR CONDITIONING WATER TREATMENT STANDARD OPERATING PROCEDURE

SOP 138-21-31

VA Palo Alto Health Care System
Palo Alto, CA 94304

Service Line:
Engineering Service

Signatory Authority:
Chief of Engineering Service

Effective Date:
December 9, 2021

Responsible Owner:
Maintenance and Repair (M&R)
Supervisor

Recertification Date:
December 31, 2026

1. PURPOSE AND AUTHORITY

This standard operating procedure (SOP) establishes guidelines to govern the water treatment of the cooling towers to prevent bacteria growth in VA Palo Alto Health Care System (VAPAHCS) open and closed loop heating, ventilation, and air conditioning (HVAC) systems. Engineering Service will comply with water treatment industry standards and maintain a high-quality level of water safety.

2. PROCEDURES

- a. Each open or closed-loop HVAC cooling tower will be tested weekly, and test results shall be recorded. Testing will be conducted and recorded for the following:
 - (1) pH level of water
 - (2) Total Dissolved Solids (TDS) of incoming tower water; and
 - (3) Testing for Organophosphonate level.
- b. The difference in readings of TDS and Organophosphonate levels will be used to calculate and adjust chemical composition of the cooling tower water.
- c. A bacteria reading will be taken and depending on the results, the microbiocide chemicals will be altered to maintain the prevention of bacteria in the water.
- d. Applicable meters will be read to document water and sewage usage levels at the facility.

- e. Chilled water, heating hot water, and condenser water closed-loops will be inspected and tested monthly. The readings will be recorded and chemicals will be added as necessary.

3. ASSIGNMENT OF RESPONSIBILITIES

- a. The Maintenance and Repair Supervisor is responsible for oversight and compliance of the monitoring requirements as required by VHA, ASHRAE, and NFPA guidelines.
- b. The Air Conditioning Shop Supervisor is responsible for the cooling tower and closed loop maintenance program. If staffing shortages exist, the supervisor will hire a contractor to perform work to maintain compliance.

4. DEFINITIONS

- a. COOLING TOWER (OPEN): The mechanical system to remove of heat from a chilled water system that passes directly through the condenser vessel and then is exposed to air when it passes through the cooling tower for evaporative cooling
- b. COOLING TOWER (CLOSED): Similar in design to the open loop tower, but the water that passes through the condenser is sealed and pressurized (closed loop), and a separate body of water passes over the tubes causing a heat exchange and thus, removal of heat.
- c. CLOSED-LOOP SYSTEM: Refers to a chilled water, heating hot water, or heat rejection loop that is sealed and pressurized and treated by use of a chemical jar or pot feeder.

5. REFERENCES

- a. ASHRAE guidelines: Minimizing the Risk of Legionellosis Associated with Building Water Systems.
- b. NFPA Standard 214: Standard on Water-Cooling Tower Latest Edition.

6. REVIEW

This SOP must be reviewed at recertification and including when there are changes to service-level privacy assessments.

7. RECERTIFICATION

This SOP is scheduled for recertification on or before the last working day of December 2026. This SOP will continue to serve as local policy until it is recertified or rescinded. In

the event of contradiction with national policy, the national policy supersedes and controls.

8. SIGNATORY AUTHORITY

JAMES P.

Digitally signed by JAMES
P. CROWE 1726175

CROWE 1726175

Date: 2021.12.10 16:45:51
-08'00'

James P. Crowe
Chief of Engineering Service

Date Approved:

NOTE: *The signature remains valid until rescinded by an appropriate administrative action.*

DISTRIBUTION: SOPs are available at

<https://dvagov.sharepoint.com/sites/PAL/memoranda/Pages/Policies.aspx>.

NOTE: This is an internal VA website that is not available to the public.

**RESCISSION OF HEALTH CARE SYSTEM MEMORANDUM (HCSM) NO. 138-18-31,
OPERATING ROOM HEATING, VENTILATION, AND AIR CONDITIONING WATER
TREATMENT POLICY**

Director's Memorandum 138-18-31R

VA Palo Alto Health Care System
Palo Alto, CA 94304

Signatory Authority:

VA Palo Alto Health Care System
Director

Rescinded Document:

HCSM No. 138-18-31, Heating,
Ventilation, and Air Conditioning water
Treatment Policy, dated September 1,
2008 (Issue August 1, 2018)

Effective Date:

December 9, 2021

1. By direction of the VA Palo Alto Health Care System (VAPAHCS) Director, 138-18-31, Heating, Ventilation, and Air Conditioning Water Treatment Policy, dated September 1, 2008, is rescinded.

2. The information has been moved to standard operating procedure. It is available at:
<https://dvagov.sharepoint.com/sites/PAL/memoranda/Pages/Policies.aspx>.

3. This Director's Memorandum will be archived as of December 2022. However, the rescission information will remain in effect.

4. SIGNATORY AUTHORITY

Lisa M. Howard
VA Palo Alto Health Care System Director

Date Approved:

NOTE: *The signature remains valid until rescinded by an appropriate administrative action.*

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